A-A-51356
7 March 1986

COMMERCIAL ITEM DESCRIPTION

BATTERY SUPPORT SYSTEM

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a battery support system capable of maintaining the batteries used in Defibrillator, Monitor, Recorder, NSN 6515-01-174-2406.

Salient characteristics:

The battery support system shall provide an integrated system to insure absolute readiness of the defibrillator/monitor system in emergency situations. The battery charger shall incorporate a microprocessor with appropriate algorithm capable of assessing a battery's energy profile and maximizing the use and maintenance of nickel-cadmium batteries only. It shall also provide routine testing of the defibrillator. The system shall be designed to reduce the development of nickel-cadmium "memory", and to remove memory by discharge/recharge cycling confirming restored capacity values. The maximum weight shall be 30 lbs.

BATTERY CHARGING

The unit shall be capable of simultaneously charging three (3) nickel-cadmium batteries to 90 percent capacity within 70 minutes at 25°C. The microprocessor shall adjust the charging rate as the battery reaches a full charged state, to eliminate overcharging. Battery status shall be provided by appropriate indicator lights, for any battery being charged. Battery charging shall be aborted by the following conditions

- The removal of a battery from the charger
- The detection of a full charge condition before overcharge occurs.
- Detection of shorted cells within a battery pack
- Detection of a low impedance non-battery load across charger terminals.

The battery charger shall provide operator cues indicating when the battery is in a main charge cycle, maintenance charge cycle or fault mode.

The charger shall respond to the monotonic voltage use time curve for a nickel cadmium battery.

AMSC N/A FSC 6625

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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If the battery is left in the charger for extended periods of time the main 1C charge shall terminate upon full charge and a .1C maintenance charge shall begin.

The charger shall be capable of delivering 1 ± 0.05 amperes average to a zero impeance load. This shall be a continuous condition for loads displaying a voltage drop between 10 and 17 volts.

Voltages below ten volts shall require a minimum of two minutes delivery.

The charge circuit shall be capable of continuous delivery of 100 ± 35 milliampere into a zero impedance load. Capability shall be continuous for all loads from 0 to 17 volts.

At several points in the algorithm a check shall be made on the absolute battery pack voltage level. If it does not agree with expected levels the charger enters a problem mode in which the charger turns off and prompts the operator.

If the battery is left in the charger for extended periods of time the topping charge shall assume a duty cycle.

BATTERY DISCHARGING AND BATTERY CAPACITY EVALUATION

The battery discharger shall be an integral part of the battery charging device. The unit shall have the capability of discharging a battery to a 1 volt per cell level to minimize the formation of "memory". The battery will then automatically be recharged. The unit shall indicate the battery capacity following a discharge cycle. The battery discharger circuit shall be provided for one battery at a time. The commanded discharge rate shall be 1 ± 0.04 amperes. Depression of the discharge key shall cause the appropriate battery to enter a discharge cycle in which the percent battery capacity is calculated. Automatic display of the percent battery capacity shall occur after the recharge cycle following the discharge of the battery. At this point a timed 1 ampere discharge of the battery shall take place. A fully charged battery pak shall be discharged in 1 hour.

DEFIBRILLATOR ENERGY EVALUATION

The unit will provide a defibrillator test load for routine testing purposes. Accuracy will be within ±5 percent.

An operator shall specify the expected defibrillator energy by scrolling the number in the display. Upon receiving a defibrillator waveform the battery charger calculates the energy level. If the energy was within ±10 percent of the scrolled-in energy, a cue shall illuminate beside the scrolled-in value. Otherwise the actual energy is displayed.

A means in which the percent battery capacity and defibrillator energy may alternately be accessed on the display is also included.

ELECTRICAL CHARACTERISTICS--The system shall be operable from 115 vac 50/60 Hz or 230 vac 50/60 Hz.

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Workmanship. The battery support system shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One battery support system, as specified, constitutes one unit.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

The following National Stock Number is covered by this document

6625-01-192-9460

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

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MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

CIVIL AGENCY COORDINATING ACTIVITY

VA-OSS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6625-0648

Located: H1-12-2

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